

An
Industrial Synopsis Report
BANK LOAN ANALYSIS USING POWER BI

Submitted in for the partial fulfilment of the degree

By

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Under the Guidance of

Faculty Internship Guide

Name: Dr. Manoj Tiwari

Industry Guide

Name: Mr. Nikhil Goklani

ABSTRACT

Power BI stands as a powerful tool in the realm of data visualization and analytics, empowering users to transform raw data into actionable insights with remarkable ease and efficiency. This study harnesses the capabilities of Power BI to conduct a comprehensive analysis of bank loans, shedding light on critical aspects of loan portfolio performance and borrower characteristics. The significance of data visualization and analytics cannot be overstated in today's data-driven world, where organizations strive to extract valuable insights from vast amounts of data to drive informed decision-making and gain a competitive edge.

Through the lens of Power BI, this study embarks on a journey to explore the intricacies of bank loan analysis, leveraging its intuitive interface and robust analytical features to delve into the complexities of loan data. By harnessing the power of interactive dashboards, advanced visualizations, and sophisticated analytics, this study endeavors to unravel key insights hidden within the vast troves of loan data, offering stakeholders a deeper understanding of loan portfolio dynamics and risk factors.

Bank loans represent a cornerstone of the financial landscape, serving as a vital source of funding for individuals and businesses alike. However, analyzing and managing loan portfolio pose significant challenges, requiring a nuanced understanding of borrower behavior, loan performance metrics, and market dynamics. By employing Power BI as a central tool in this analysis, this study seeks to bridge the gap between raw loan data and actionable insights, enabling stakeholders to make informed decisions, mitigate risks, and optimize portfolio performance in the ever-evolving financial landscape.

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Internship Organization Details

Bewesocial is a leading digital marketing agency and social media agency based in Jodhpur, India. Established in 2020, we specialize in providing comprehensive digital marketing solutions to businesses of all sizes. Our team of experts is dedicated to helping clients achieve their marketing goals through innovative strategies and effective campaigns. In addition to digital marketing services, we also offer analytics services using advanced tools such as Power BI and Tableau to provide valuable insights and drive data-driven decisions. At Bewesocial, we are committed to delivering exceptional results and maximizing the online presence and visibility of our clients.

Industry: Advertising Services

Services provided:

1. Digital Marketing Services:

- a. Social Media Marketing
- b. Search Engine Optimization (SEO)
- c. Pay-Per-Click (PPC) Advertising
- d. Content Marketing
- e. Email Marketing
- f. Online Reputation Management (ORM)
- g. Influencer Marketing

2. Analytics Services:

- a. Power BI
- b. Tableau
- c. Data Visualization
- d. Data Analysis
- e. Business Intelligence

Industry Mentor Profile

Name: Mr. Nikhil Goklani

Designation: Founder

Experience: 4+ years

Technical Expertise:

- Project Management
- Digital Marketer
- Data Analysis Tools
- SQL, Python
- Web Development

Achievements:

- Built a unified team by creating an agency BeWeSocial.com
- Volunteered as a Social Worker in Robin Hood Army organization.

Internship Profile

Profile Overview:

As a Software Developer Intern, you will build dynamic and engaging websites using HTML, CSS, and JavaScript. You will learn about various data analytics tools such as Power BI. By harnessing the power of data visualization and analysis, you have to uncover valuable insights and drive informed decision-making processes. You have to manage and manipulate large datasets and able to design and optimize database structures, execute complex queries and ensure data integrity and security.

Responsibilities:

- Collaborate with cross-functional teams to develop and implement responsive and user-friendly web interfaces using HTML, CSS, and JavaScript.
- Assist in the design and development of database structures using SQL, ensuring efficient data storage and retrieval processes.
- Gain hands-on experience in building dynamic and interactive websites, incorporating best practices in front-end development to enhance user experience.
- Learn and apply data analytics techniques using tools such as Power BI to visualize and analyze large datasets, extracting valuable insights to inform decision-making processes.
- Participate in code reviews and debugging sessions to identify and resolve issues, ensuring the quality and performance of web applications.
- Continuously expand knowledge and skills in web development, data analytics, and database management through self-directed learning and mentorship opportunities.
- Contribute to the development and maintenance of software documentation, including technical specifications, user manuals, and code comments, to support team collaboration and knowledge sharing.

Technical Skills:

- Experience working with HTML, CSS, JavaScript.
- Familiarity with Bootstrap, SpringBoot, VSCode.
- Fluent Hindi and English, both Verbal and Written.
- Mastery of SQL

- Knowledge of at least one component: Data Cleaning, Data Gathering, KPIs, Power BI, Dashboards.
- Experience with any programming language used for data engineering purpose: Python, Java, R, etc.
- Ability to make visually appealing and user-friendly interfaces that enhance the overall user experience.
- Curiosity and eagerness to learn or explore the realm of data analytics to gain proficiency in interpreting data and extracting meaningful conclusions.
- Can apply technical expertise to solve real-world problems and make a positive impact.

MODE OF INTERNSHIP – OFFLINE

LOCATION – 484, SARDARPURA B ROAD, JODHPUR RAJASTHAN 342003

Literature Overview (Technology Used)

Web Development:

Web development encompasses the process of creating, building, and maintaining websites and web applications that are accessible via the internet. It involves a combination of various skills, technologies, and methodologies to produce functional and visually appealing digital experiences for users. Web development can be broadly categorized into two main areas: front-end development and back-end development.

Front-end development focuses on the client-side of web applications, dealing with the design, layout, and functionality that users interact with directly in their web browsers. Front-end developers use languages such as HTML (Hypertext Markup Language), CSS (Cascading Style Sheets), and JavaScript to create interactive and responsive user interfaces. They are responsible for ensuring that websites are visually engaging, easy to navigate, and optimized for different devices and screen sizes.

Back-end development, on the other hand, involves the server-side of web applications, dealing with the logic, databases, and server configurations that power the front-end functionality. Back-end developers work with programming languages such as PHP, Python, Ruby, Java, and databases like MySQL, PostgreSQL, and MongoDB to create the server-side infrastructure that supports the functionality of web applications. They focus on tasks such as database management, server configuration, authentication, and data processing.

Technology related to Web Development:

Web development is closely intertwined with various technologies and tools that enable developers to create, manage, and optimize websites and web applications. Some of the key technologies related to web development include:

1. **HTML (Hypertext Markup Language):** HTML is the standard markup language used to create the structure and content of web pages. It provides the foundation for building web documents and defines the layout, text, images, links, and other elements within a webpage.
2. **CSS (Cascading Style Sheets):** CSS is a stylesheet language used to define the presentation and styling of HTML elements on a webpage. It allows developers to control

the appearance, layout, colours, fonts, and other visual aspects of web content, enhancing the design and user experience.

3. **JavaScript:** JavaScript is a dynamic scripting language that enables interactive and dynamic functionality on web pages. It is commonly used for client-side scripting to add interactivity, validate forms, manipulate HTML elements, create animations, and handle user events.
4. **Databases:** Databases such as MySQL, PostgreSQL, MongoDB, and SQLite are used to store, retrieve, and manage data for web applications. SQL stands for Structure Query Language and it is a standard Database Language that is used to create, maintain, and retrieve the relational. Overall, these technologies form the foundation of modern web development, empowering developers to create dynamic, interactive, and scalable web applications that meet the evolving needs of users and businesses on the internet.

Data Analysis:

Data analysis is the process of examining, cleaning, transforming, and interpreting data to uncover meaningful insights, patterns, and trends that can inform decision-making and drive business outcomes. It involves a combination of statistical, mathematical, and computational techniques to extract actionable information from raw data and derive valuable insights that can be used to optimize processes, improve performance, and gain a competitive advantage.

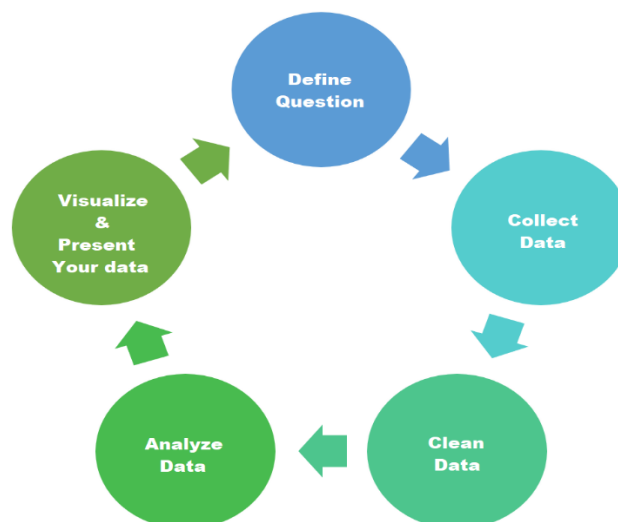


Fig. 4.3. Data Analysis

Data Collection Process

The data collection process involved gathering loan data from multiple sources to ensure a comprehensive analysis. The primary sources of data included:

- **Internal Databases:** Loan data from the bank's internal databases provided detailed information about borrower demographics, loan terms, and repayment history.
- **External Datasets:** Publicly available datasets and third-party data providers were utilized to supplement internal data and enhance the breadth of analysis.
- **APIs:** Application Programming Interfaces (APIs) were used to access real-time financial data and economic indicators for contextual analysis.

The data collection process adhered to ethical guidelines and data privacy regulations to ensure the confidentiality and integrity of borrower information.

Data Preprocessing, Cleaning, and Transformation

Before conducting any analysis, the collected data underwent a series of preprocessing, cleaning, and transformation steps to ensure accuracy and consistency:

1. Data Cleaning:

- **Removal of Duplicates:** Duplicate records were identified and removed to eliminate redundancy in the dataset.
- **Handling Missing Values:** Missing values were imputed using appropriate techniques such as mean imputation or predictive modelling to preserve data integrity.
- **Outlier Detection:** Outliers were identified and treated to prevent them from skewing the analysis results.

2. Data Integration:

- **Harmonization:** Integrate data from diverse sources, standardizing variables, units, and formats for consistency.

- **Unification:** Merge datasets to create a cohesive dataset, ensuring seamless analysis across all sources.
- **Standardization:** Align variable names and data structures, streamlining integration processes and enhancing dataset usability.

3. Feature Engineering:

- **Insight Generation:** Derive new variables to capture additional insights, enhancing predictive modelling performance.
- **Complexity Reduction:** Aggregate data to simplify analysis and improve model interpretability.
- **Transformation:** Apply mathematical functions to variables, transforming data to meet specific analytical requirements and enhance model accuracy.

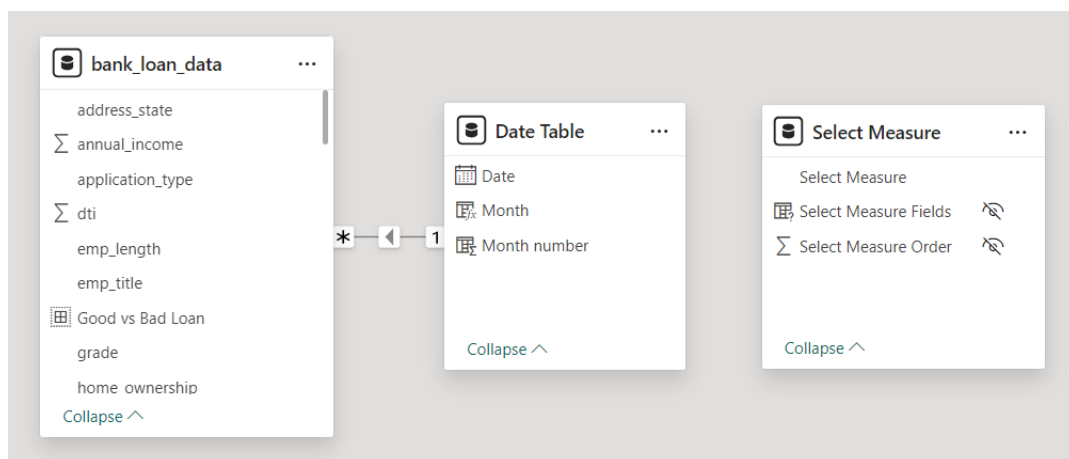


Fig. 4.3. Model View

4. Data Transformation:

- **Standardization:** Numerical variables were standardized to a common scale to facilitate comparison and analysis.
- **Encoding Categorical Variables:** Categorical variables were encoded into numerical format using techniques like one-hot encoding or label encoding for compatibility with analytical models.

- **Feature Engineering:** New features were created from existing variables to capture additional insights relevant to loan analysis, such as debt-to-income ratios or credit utilization metrics.

Analytical Techniques for Bank Loan Analysis

Bank loan analysis relies on a suite of analytical techniques to evaluate various aspects of loan portfolios, borrower creditworthiness, and risk management strategies. These techniques serve diverse purposes, ranging from summarizing loan characteristics to forecasting default probabilities and optimizing portfolio allocation.:

1. **Descriptive Analysis:** Descriptive statistics like mean, median, and standard deviation are employed to summarize loan portfolios, borrower profiles, and performance metrics.
2. **Predictive Modelling:** Techniques such as logistic regression, decision trees, and ensemble methods are utilized to forecast default probabilities, estimate credit risk, and evaluate borrower creditworthiness.
3. **Segmentation Analysis:** Segmentation techniques like cluster analysis and decision trees help categorize borrowers into distinct groups based on common characteristics, enabling tailored lending products and risk management strategies.
4. **Financial Ratio Analysis:** Calculation of key financial ratios such as Debt-to-Income Ratio and Debt Service Coverage Ratio aids in assessing borrower financial health and repayment capacity.
5. **Time-Series Analysis:** Analyzing trends and patterns in loan performance metrics over time using techniques like moving averages and ARIMA models assists in identifying cyclical trends and predicting future loan demand.
6. **Probability of Default (PD) Modelling:** Survival analysis and advanced machine learning algorithms like Support Vector Machines and Neural Networks are utilized to model the probability of default over time.
7. **Portfolio Optimization:** Risk segmentation and portfolio diversification strategies are employed to optimize allocation of resources and minimize concentration risk, thereby enhancing portfolio stability.

By employing these methodologies and techniques, we were able to conduct comprehensive bank loan analysis, identify key risk factors, and optimize lending strategies to achieve desired business outcomes.

Database (SQL):

SQL (Structured Query Language) is a powerful and widely used programming language for managing and manipulating relational databases. It provides a standardized syntax and set of commands for querying, updating, and managing data stored in relational database management systems (RDBMS).

Here are some key aspects and features of SQL:

1. **Data Querying:** SQL allows users to retrieve data from databases using the SELECT statement. Users can specify which columns to retrieve, filter rows based on certain criteria using the WHERE clause, sort the results using the ORDER BY clause, and group data using the GROUP BY clause. SQL also supports aggregate functions such as SUM, AVG, COUNT, MIN, and MAX for performing calculations on data.
2. **Data Manipulation:** SQL provides commands for adding, updating, and deleting data in relational databases. The INSERT statement is used to add new records to a table, the UPDATE statement is used to modify existing records, and the DELETE statement is used to remove records from a table.
3. **Data Definition:** SQL allows users to define the structure and organization of databases and tables using Data Definition Language (DDL) statements. DDL statements include commands such as CREATE TABLE, ALTER TABLE, and DROP TABLE for creating, modifying, and deleting tables, as well as commands for defining constraints, indexes, and relationships between tables.
4. **Data Control:** SQL provides commands for managing user access and permissions to databases and tables. Data Control Language (DCL) statements such as GRANT and REVOKE are used to grant or revoke privileges to users and roles, allowing administrators to control who can access and manipulate data in the database.
5. **Data Integrity:** SQL supports various constraints and integrity rules to ensure the accuracy, consistency, and reliability of data stored in databases. Constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK are used to enforce data integrity rules and prevent invalid or inconsistent data from being entered into tables.

6. **Transaction Control:** SQL provides commands for managing transactions, which are sequences of database operations that are executed as a single unit of work. Transaction Control Language (TCL) statements such as COMMIT, ROLLBACK, and SAVEPOINT are used to control the atomicity, consistency, isolation, and durability (ACID) properties of transactions.

SQL is a versatile and powerful language that is used by database administrators, data analysts, software developers, and other IT professionals to interact with and manipulate data in relational databases. It is supported by a wide range of relational database management systems (RDBMS) such as MySQL, PostgreSQL, Oracle, SQL Server, and SQLite, making it a fundamental skill for anyone working with data-driven applications and systems.

Power BI:

Power BI is a business intelligence and data visualization tool developed by Microsoft. It enables users to analyse and visualize data from various sources, create interactive reports and dashboards, and share insights with stakeholders. Power BI is part of the Microsoft Power Platform, which also includes Power Apps and Power Automate (formerly known as Microsoft Flow).

Here are some key features and capabilities of Power BI:

1. **Data Connectivity:** Power BI supports connectivity to a wide range of data sources, including databases, cloud services, files, and streaming data. Users can connect to sources such as SQL Server, Azure SQL Database, Excel, SharePoint, Dynamics 365, Salesforce, Google Analytics, and more.
2. **Data Preparation:** Power BI provides tools for cleaning, transforming, and shaping data to prepare it for analysis. The Power Query Editor allows users to perform data cleaning, filtering, merging, and other transformations using a graphical interface. Users can create custom data queries and apply transformations to clean and reshape data before loading it into Power BI.
3. **Data Modeling:** Power BI includes a powerful data modeling engine that allows users to define relationships between data tables, create calculated columns and measures, and build complex data models. Users can define hierarchies, create calculated fields using DAX (Data Analysis Expressions), and optimize data models for performance.

4. **Data Visualization:** Power BI enables users to create interactive and visually appealing reports and dashboards to visualize insights from their data. Users can design custom visuals, charts, graphs, maps, and tables using a drag-and-drop interface. Power BI offers a wide range of visualization options and formatting capabilities to customize the appearance of reports and dashboards.
5. **Collaboration and Sharing:** Power BI allows users to collaborate on reports and dashboards by sharing them with colleagues and stakeholders. Users can publish reports to the Power BI service, share them via web links, embed them in websites and applications, and distribute them via email or mobile devices. Power BI also supports integration with Microsoft Teams, SharePoint, and other collaboration tools.
6. **AI and Advanced Analytics:** Power BI integrates with Azure AI services to enable advanced analytics capabilities such as predictive modeling, anomaly detection, sentiment analysis, and natural language processing. Users can leverage AI-powered features such as AI visuals, Q&A (natural language querying), and machine learning models to gain deeper insights from their data.

Overall, Power BI empowers users to turn data into actionable insights and make informed decisions by providing powerful tools for data analysis, visualization, and collaboration. It is widely used by businesses and organizations of all sizes to drive data-driven decision-making and gain competitive advantage in today's digital world.

Project Overview

Bank Loan Analysis Using Power BI

Domain Knowledge

Bank loans are a crucial financial tool that enables individuals and businesses to achieve their goals and manage financial needs. However, it's essential for borrowers to understand the terms, costs, and responsibilities associated with loans to make informed financial decisions.

Banks collect loan data through various channels and processes, including:

- **Loan Applications:** When individuals or businesses apply for loans, they submit detailed applications that include personal and financial information. This data is collected electronically or in paper form.
- **Credit Reports:** Banks often access credit reports from credit bureaus when assessing a borrower's creditworthiness. These reports contain information about a person's credit history, existing loans, and payment behaviour.
- **Internal Records:** Banks maintain internal records of loan transactions, including disbursements, repayments, and loan status changes. These records are generated and stored in the bank's database.
- **Online Portals:** Many banks offer online platforms where borrowers can apply for loans, make payments, and access account information. Data from these portals is collected and stored for analysis.
- **Third-party Data Sources:** Some banks may use external data sources, such as income verification services, to gather additional information about borrowers.

Process of Granting a Loan

- **Loan Application:** The process begins when a customer submits a loan application to a bank or lending institution. This application can be submitted in person, online, or through other channels.
- **Application Review:** The lending institution reviews the loan application and collects necessary documentation, such as income statements, credit reports, and identification documents.
- **Identity Verification:** One of the initial checks is to verify the applicant's identity. This helps ensure that the applicant is who they claim to be and prevents identity theft.

- **Credit Check:** A crucial step is to perform a credit check on the applicant. This involves accessing their credit report from credit bureaus. Lenders evaluate the applicant's credit history, credit score, and any past delinquencies or defaults.
- **Income Verification:** Lenders assess the applicant's ability to repay the loan by verifying their income. This may involve reviewing pay stubs, tax returns, or other income documentation.
- **Debt-to-Income Ratio (DTI) Check:** Lenders calculate the applicant's DTI, which is the ratio of their monthly debt payments to their monthly income. A lower DTI indicates better repayment capacity.
- **Employment Verification:** Lenders may contact the applicant's employer to verify their employment status and length of employment. Stable employment history is often seen as a positive factor.
- **Collateral Assessment (if applicable):** If the loan is secured by collateral, such as a home or a car, the lender evaluates the value and condition of the collateral.
- **Risk Assessment:** Lenders assess the overall risk associated with the loan. This includes considering the applicant's credit risk, income stability, and the purpose of the loan.
- **Loan Approval or Denial:** Based on the information gathered and the risk assessment, the lender makes a decision to approve or deny the loan application. If approved, the lender determines the loan amount, interest rate, and terms.
- **Loan Agreement:** If the loan is approved, the lender provides the applicant with a loan agreement that outlines the terms and conditions, including the interest rate, repayment schedule, and any fees.
- **Disbursement of Funds:** Once the loan agreement is signed by both parties, the lender disburses the funds to the borrower. The borrower can use the funds for the specified purpose.
- **Repayment:** The borrower is responsible for making regular loan payments as specified in the loan agreement. This includes repaying the principal amount along with interest.
- **Ongoing Monitoring:** Lenders continue to monitor the loan throughout its term, including tracking payments, assessing the borrower's financial health, and managing any delinquencies or defaults.

Reasons for Analysing Bank Loan Data:

Banks analyse loan data for several critical reasons:

- **Risk Assessment:** One of the primary purposes of analysing loan data is to assess the risk associated with lending to a particular individual or business. Banks use data to evaluate the creditworthiness of borrowers, predict default probabilities, and determine interest rates and lending terms.
- **Decision-making:** Loan data analysis supports the decision-making process when evaluating loan applications. Banks use data-driven models and algorithms to make informed lending decisions, such as approving or denying loan requests.
- **Portfolio Management:** Banks manage portfolios of loans, including mortgages, personal loans, and business loans. Data analysis helps banks monitor the health of these portfolios, identify underperforming loans, and optimize loan terms and pricing.
- **Fraud Detection:** Banks use data analysis to detect fraudulent loan applications and activities. Unusual patterns, inconsistencies, or discrepancies in loan data can trigger fraud alerts.
- **Regulatory Compliance:** Banks are subject to regulatory requirements that mandate the collection and reporting of loan data. Compliance with regulations such as the Home Mortgage Disclosure Act (HMDA) and the Know Your Customer (KYC) regulations requires data analysis and reporting.
- **Customer Insights:** Analysing loan data provides insights into customer behaviour, preferences, and needs. Banks can use these insights to tailor loan products and marketing strategies to specific customer segments.
- **Profitability Analysis:** Banks assess the profitability of their loan portfolios by analysing data related to interest income, loan origination costs, default rates, and collection efforts.
- **Market Research:** Data analysis helps banks understand market trends, competitive landscape, and customer demand. This information guides product development and market expansion strategies.
- **Credit Risk Management:** Banks continuously monitor and manage credit risk associated with their loans. Data analysis helps in setting risk management strategies, provisioning for potential losses, and stress testing loan portfolios.
- **Customer Retention:** Banks use data analysis to identify opportunities for retaining existing customers, such as offering loan refinancing options or additional financial products.

Terminologies used in Data

Fields Used in Data

- **Loan ID:**

Purpose: Loan ID is a unique identifier assigned to each loan application or loan account. It serves as a primary key for tracking and managing individual loans.

Use for Banks: Banks use Loan IDs to efficiently manage and track loans throughout their lifecycle. It aids in organizing loan records, monitoring repayments, and addressing customer inquiries.

- **Address State:**

Purpose: Address State indicates the borrower's location. It helps in assessing regional risk factors, compliance with state regulations, and estimating default probabilities.

Use for Banks: Banks use this information to identify regional trends in loan demand, adjust marketing strategies, and manage risk portfolios based on geographic regions.

- **Employee Length:**

Purpose: Employee Length provides insights into the borrower's employment stability. Longer employment durations may indicate greater job security.

Use for Banks: Banks consider employment length when assessing a borrower's ability to repay. Stable employment often translates to a lower default risk.

- **Employee Title:**

Purpose: Employee Title specifies the borrower's occupation or job title. It helps lenders understand the source of the borrower's income.

Use for Banks: Banks use this field to verify income sources, assess the borrower's financial capacity, and tailor loan offers to different professions.

- **Grade:**

Purpose: Grade represents a risk classification assigned to the loan based on creditworthiness. Higher grades signify lower risk.

Use for Banks: Banks use the grade to price loans and manage risk. Higher-grade loans typically receive lower interest rates and are more attractive to investors.

- **Sub Grade:**

Purpose: Sub Grade refines the risk assessment within a grade, providing additional risk differentiation.

Use for Banks: Sub Grades offer a finer level of risk assessment, helping banks tailor interest rates and lending terms to match borrower risk profiles.

- **Home Ownership:**

Purpose: Home Ownership indicates the borrower's housing status. It offers insights into financial stability.

Use for Banks: Banks use this field to assess collateral availability and borrower stability. Homeowners may have lower default rates.

- **Issue Date:**

Purpose: Issue Date marks the loan's origination date. It's crucial for loan tracking and maturity calculations.

Use for Banks: Banks use Issue Dates to track loan aging, calculate interest accruals, and manage loan portfolios.

- **Last Credit Pull Date:**

Purpose: Last Credit Pull Date records when the borrower's credit report was last accessed. It helps monitor creditworthiness.

Use for Banks: Banks use this date to track credit history updates, assess credit risk, and make informed lending decisions.

- **Last Payment Date:**

Purpose: Last Payment Date marks the most recent loan payment received. It tracks the borrower's payment history.

Use for Banks: Banks use this date to assess payment behavior, calculate delinquency, and project future payments.

- **Loan Status:**

Purpose: Loan Status indicates the current state of the loan (e.g., fully paid, current, default). It tracks loan performance.

Use for Banks: Banks use Loan Status to monitor loan health, categorize loans for risk analysis, and determine provisioning requirements.

- **Next Payment Date:**

Purpose: Next Payment Date estimates the date of the next loan payment. It assists in cash flow forecasting.

Use for Banks: Banks use this date for liquidity planning and to project revenue from loan portfolios.

- **Purpose:**

Purpose: Purpose specifies the reason for the loan (e.g., debt consolidation, education). It helps understand borrower intentions.

Use for Banks: Banks use this field to segment and customize loan offerings, aligning loan terms with borrower needs.

- **Term:**

Purpose: Term defines the duration of the loan in months. It sets the repayment period.

Use for Banks: Banks use the term to structure loan agreements, calculate interest payments, and manage loan maturities.

- **Verification Status:**

Purpose: Verification Status indicates whether the borrower's financial information has been verified. It assesses data accuracy.

Use for Banks: Banks use this field to gauge data reliability, verify income, and evaluate loan application credibility.

- **Annual Income:**

Purpose: Annual Income reflects the borrower's total yearly earnings. It assesses repayment capacity.

Use for Banks: Banks use this income figure to determine loan eligibility, calculate debt-to-income ratios, and evaluate creditworthiness.

- **DTI (Debt-to-Income Ratio):**

Purpose: DTI measures the borrower's debt burden relative to income. It gauges the borrower's capacity to take on additional debt.

Use for Banks: Banks use DTI to assess a borrower's ability to handle loan payments and make responsible lending decisions.

- **Instalment:**

Purpose: Instalment is the fixed monthly payment amount for loan repayment, including principal and interest.

Use for Banks: Banks use this field to structure loan terms, calculate amortization schedules, and assess payment affordability.

- **Interest Rate:**

Purpose: Interest Rate represents the annual cost of borrowing expressed as a percentage. It determines the loan's cost.

Use for Banks: Banks use interest rates to price loans, manage profit margins, and attract investors.

- **Loan Amount:**

Purpose: Loan Amount is the total borrowed sum. It defines the principal amount.

Use for Banks: Banks use Loan Amount to determine loan size.

Key Performance Indicators (KPIs)

KPIs, or Key Performance Indicators, are measurable metrics that organizations use to evaluate and track the success of their business objectives and strategic initiatives. KPIs are quantifiable, specific, and relevant to the organization's goals, providing actionable insights into performance and progress over time.

Here are some key aspects and characteristics of KPIs:

- 1. Alignment with Business Objectives:** KPIs should be closely aligned with the organization's overall mission, vision, and strategic goals. They should reflect the critical success factors that drive the organization's performance and success.
- 2. Measurable and Quantifiable:** KPIs should be measurable using objective data and metrics. They should provide clear and specific indicators of performance, allowing stakeholders to track progress and make data-driven decisions.
- 3. Relevance and Meaningful:** KPIs should be relevant to the specific objectives and priorities of the organization. They should measure aspects of performance that are critical to achieving desired outcomes and provide meaningful insights into performance drivers.
- 4. Actionable and Insightful:** KPIs should provide actionable insights that can inform decision-making and drive improvements. They should highlight areas of strength and weakness, identify trends and patterns, and prompt corrective actions or strategic adjustments as needed.
- 5. Timely and Real-Time Monitoring:** KPIs should be monitored and tracked regularly to ensure timely feedback and intervention. Real-time monitoring allows organizations to identify issues and opportunities quickly, enabling proactive management and response.
- 6. Balanced and Comprehensive:** KPIs should provide a balanced view of performance across different dimensions of the organization, such as financial, operational, customer, and employee perspectives. A balanced set of KPIs ensures that all aspects of performance are considered and managed effectively.

7. SMART Criteria: KPIs should adhere to the SMART criteria – Specific, Measurable, Achievable, Relevant, and Time-bound. This ensures that KPIs are well-defined, attainable, and meaningful for driving performance improvement.

Examples of KPIs vary depending on the organization and industry but may include metrics such as revenue growth, customer satisfaction, employee productivity, operational efficiency, market share, profitability, and more. KPIs are typically tracked and monitored using dashboards, scorecards, and performance reports, allowing stakeholders to visualize performance trends and make informed decisions based on data-driven insights.

Overall, KPIs play a crucial role in helping organizations measure progress, evaluate performance, and drive continuous improvement toward achieving their strategic objectives and business goals.

Dashboards

Power BI dashboards are interactive data visualization tools that allow users to monitor, analyze, and visualize key metrics and insights from their data in real-time. Power BI dashboards provide a consolidated view of important KPIs (Key Performance Indicators), trends, and patterns, enabling users to make informed decisions and take timely action based on actionable insights.

Here are some key features and characteristics of Power BI dashboards:

- 1. Interactive Data Visualization:** Power BI dashboards enable users to create visually appealing and interactive visualizations of their data, including charts, graphs, maps, tables, and KPI tiles. Users can customize the appearance and layout of dashboard elements to highlight important insights and trends.
- 2. Real-Time Data Updates:** Power BI dashboards can connect to live data sources and automatically update visualizations in real-time as new data becomes available. This allows users to monitor key metrics and trends as they happen and react quickly to changes or anomalies.
- 3. Customizable Layouts:** Power BI dashboards offer flexible layout options, allowing users to arrange and organize visualizations to suit their preferences and requirements. Users can resize, move, and group dashboard elements to create a personalized view of their data.
- 4. Drill-Down and Drill-Through:** Power BI dashboards support drill-down and drill-through capabilities, allowing users to explore data at different levels of detail and navigate between

related visualizations. This enables users to uncover insights and understand the underlying factors driving performance.

5. Cross-Filtering and Interactivity: Power BI dashboards support cross-filtering and interactivity between visualizations, enabling users to dynamically interact with data and explore relationships between different metrics and dimensions. Users can click on data points to filter other visualizations and gain deeper insights into their data.

6. Sharing and Collaboration: Power BI dashboards can be shared with colleagues and stakeholders within an organization or externally with clients and partners. Users can publish dashboards to the Power BI service, share them via web links or embedded URLs, and collaborate on dashboard design and development.

7. Mobile Access: Power BI dashboards are optimized for mobile devices, allowing users to access and view dashboards on smartphones and tablets using the Power BI mobile app. Mobile users can interact with visualizations, explore data, and receive notifications and alerts on the go.

Overall, Power BI dashboards provide a powerful and flexible platform for visualizing and analyzing data, enabling users to gain actionable insights, monitor performance, and drive decision-making across their organization. With its intuitive interface, real-time updates, and interactive features, Power BI dashboards empower users to unlock the full potential of their data and drive business success.

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